



MYASTHENIA GRAVIS (MG)



PATHOPHYSIOLOGY

- Myasthenia Gravis is a chronic autoimmune neuromuscular disorder where antibodies attack acetylcholine receptors (AChR) at the neuromuscular junction (the point where nerves communicate with muscles), leading to skeletal muscle weakness.
- It worsens with activity (fatigue) and improves with rest.



KEY SYMPTOMS

- Muscle weakness that improves with rest
- Ptosis (drooping eyelids)
- Diplopia (double vision)
- Dysphagia (difficulty swallowing)
- Dysarthria (slurred speech)
- Weak neck and limb muscles
- Respiratory compromise if diaphragmatic muscles are involved → Myasthenic Crisis



DIAGNOSIS

- Tensilon (edrophonium) test: temporary improvement of muscle strength = confirms diagnosis
 - ⚠️ Have atropine ready (antidote) due to risk of bradycardia
- Anti-AChR antibody blood test: positive in most patients
- Electromyography (EMG): detects abnormal muscle response to nerve stimulation
- CT/MRI of chest: to check for thymoma (tumor of the thymus gland)



TREATMENT

- Pyridostigmine (Mestinon): cholinesterase inhibitor; increases ACh at neuromuscular junction
 - 🕒 Must be given on time to avoid muscle weakness
- Corticosteroids: reduce autoimmune response
- Immunosuppressants: e.g., azathioprine

- IVIG or plasmapheresis: for crisis or severe exacerbation
- Thymectomy: surgical removal of thymus may improve symptoms



MYASTHENIC CRISIS VS. CHOLINERGIC CRISIS

Feature	Myasthenic Crisis	Cholinergic Crisis
Cause	Undermedication	Overmedication
Signs	Severe weakness, resp. failure	Weakness + salivation, sweating, bradycardia, diarrhea
Treatment	↑ pyridostigmine	Stop meds + give atropine



Tensilon test helps differentiate:



HOW IT WORKS:

- Edrophonium (Tensilon) is a short-acting acetylcholinesterase inhibitor.
- It temporarily increases acetylcholine at the neuromuscular junction.
- A small IV dose (usually 2 mg test dose, then 8 mg) is administered under observation.
 - Symptoms improve = Myasthenic Crisis
 - Symptoms worsen = Cholinergic Crisis



NURSING CARE

- Prioritize airway management and monitor for respiratory fatigue
- Administer pyridostigmine exactly on time (risk of crisis if missed)
- Schedule meds before meals to aid in swallowing
- Promote rest between activities
- Prevent aspiration: upright position for meals, suction ready
- Avoid stress, infections, temperature extremes (can worsen symptoms)



PATIENT EDUCATION

- Avoid medications that worsen MG: beta blockers, aminoglycosides, sedatives
- Encourage vaccination (avoid infections)

- Wear medical alert bracelet
- Educate about recognizing early signs of crisis and when to seek help



NCLEX HIGH-YIELD SUMMARY

Topic	Details
Med of choice	Pyridostigmine (on time!)
Emergency	Myasthenic Crisis = respiratory support
Diagnostic	Tensilon test, AChR antibodies
Cholinergic Crisis	Atropine = antidote
Nursing focus	Airway, aspiration risk, rest scheduling